

Simple Robotic Dog – Initial Mechanical Design

Task: Preliminary mechanical design of a simple quadruped robot dog (Fusion 360 model: "dogrobot")

1. Body & Frame Shape

Rectangular box chassis for the torso, with a smaller box for the head mounted on top of the front section. Cone-shaped snout and ears are decorative/sensor-mount features, not structural. This flat-panel construction keeps the design simple to manufacture (laser-cut acrylic/aluminum panels or 3D-printed shell) and gives a large, flat internal volume for batteries, controller board, and wiring.

2. Leg Design

Four straight, single-segment rigid legs (no knee joint) — the simplest configuration for a first prototype. Each leg connects to the body through one revolute (hinge) joint at the hip. This is intentionally minimal: a single-segment leg is easier to control and build than a multi-link leg, at the cost of a shorter effective stride and less adaptability to uneven terrain.

3. Number of Joints & Degrees of Freedom

4 joints total — one revolute hip joint per leg, each with 1 DOF (rotation in the sagittal/forward-back plane). Total system DOF = 4. No ankle or knee joints in this version, which is consistent with a basic, non-advanced prototype.

4. Motor Selection

Standard hobby digital servo motors are the practical choice at this scale (light weight, built-in position control, no separate driver/encoder needed). Based on the torque estimate below, a servo in the 15–20 kg·cm range is recommended, for example a DS3218 (20 kg·cm) or similar metal-gear digital servo. A common choice like the MG996R (~9.4–11 kg·cm) is undersized for this load with no safety margin, so it should be avoided unless the robot's weight is reduced.

5. Initial Torque Calculation (One Hip Joint)

Assumptions (to be confirmed once material and final dimensions are set): estimated total robot mass 1.8 kg; worst case is the robot supported on 3 legs during a walking step (1 leg lifted); leg length (hip to foot) 0.12 m; safety factor 1.5.

Step	Calculation	Result
Weight per supporting leg	$(1.8 \text{ kg} \times 9.81 \text{ m/s}^2) / 3 \text{ legs}$	5.89 N
Required torque (no margin)	$5.89 \text{ N} \times 0.12 \text{ m}$	0.71 N·m ($\approx 7.2 \text{ kg}\cdot\text{cm}$)
Required torque (with 1.5x safety factor)	$0.71 \text{ N}\cdot\text{m} \times 1.5$	1.06 N·m ($\approx 10.8 \text{ kg}\cdot\text{cm}$)

Result: a minimum of ~10.8 kg·cm torque per hip joint is required. A servo rated 15–20 kg·cm is recommended to keep the motor comfortably below its stall torque during normal operation.

6. Stability & Center of Gravity

With the head and battery mounted toward the front-top of the body, the center of gravity sits above and slightly forward of the geometric center of the 4-leg support base. Stability is acceptable in a static stance (all 4 legs down), but marginal once one leg lifts for walking — the support polygon shrinks to a triangle, so heavier components (battery, controller) should be kept low and centered to reduce tipping risk.

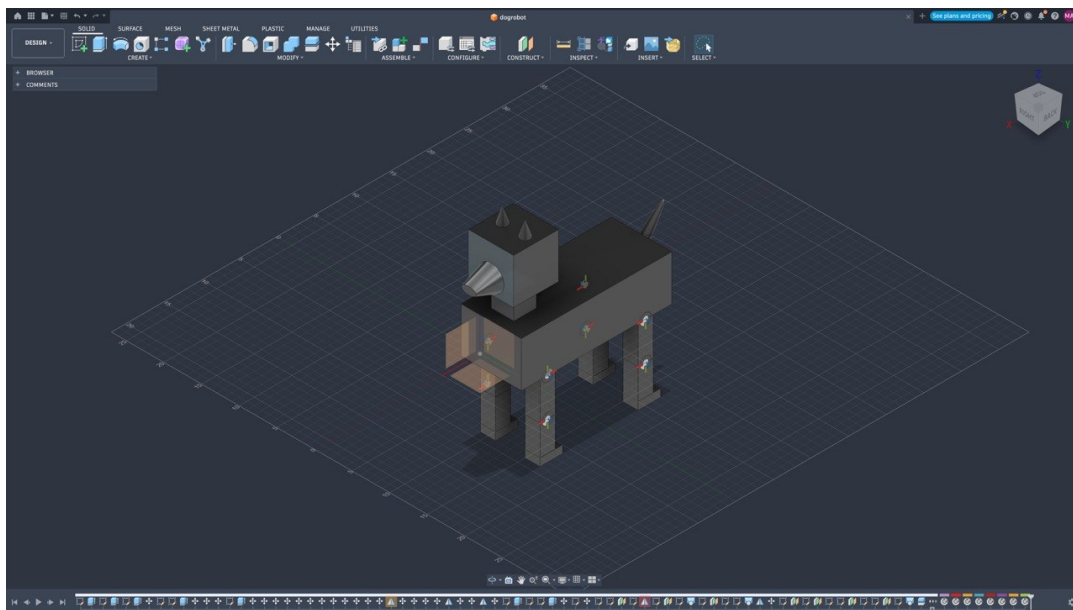
7. Proposed Walking Method

Static (crawl) gait: only one leg lifts and moves at a time while the other three remain on the ground, keeping the center of gravity inside the support triangle at all times. This is the safest and simplest gait for a first prototype with single-DOF legs and no active balance control. A trot gait (diagonal leg pairs moving together) is faster but requires better dynamic balance and is not recommended until the mechanical base is validated.

8. Expected Mechanical Problems

Single-segment legs give a short stride and cannot adapt well to uneven surfaces (no knee/ankle compliance). Servo torque margin is tight — any added weight (wiring, extra electronics) can push required torque above the motor rating. Support triangle during 3-leg stance is a stability risk, especially on non-level ground. Rigid hip-only joints will produce a stiff, non-organic walking motion. 3D-printed or thin-panel legs may flex or crack under repeated load cycles if wall thickness is too low. No ankle joint means the foot cannot stay flat on the ground through the full step — footpads should be rounded or use a passive compliant tip.

Fusion 360 Model



Working file (Fusion 360, Autodesk cloud): <https://a360.co/4yg5Wt8>